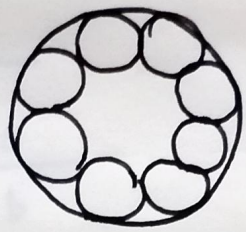
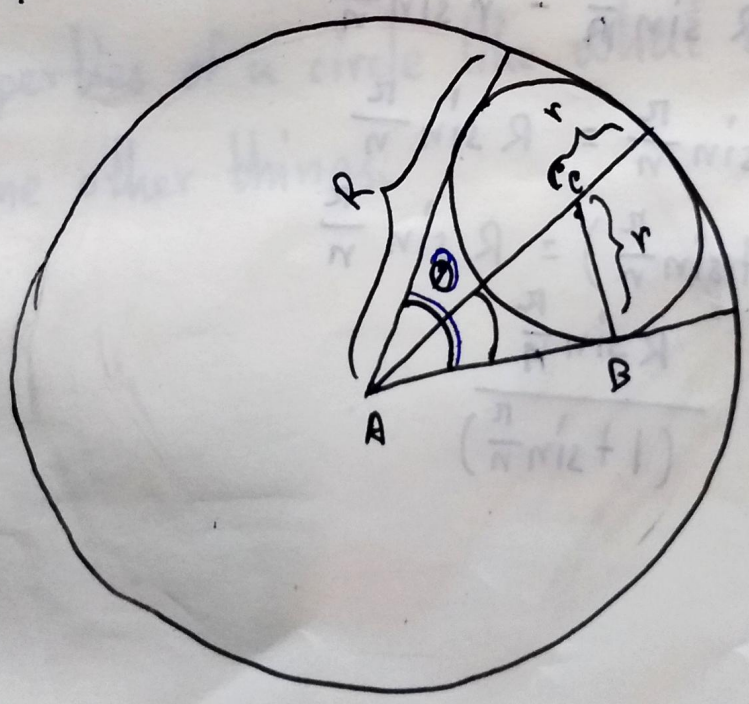
Problem 6: There is a large circle with radius R and n little circles each having radius r, are placed inside on the border of the large circle. There are no space between the circles. Given R and n, compute r. (Light OJ. 1072)



Solution:



Each small circle creates θ within the circle.

$$\therefore n \times \theta = 2\pi \rightarrow \theta = \frac{2\pi}{n}$$

$$\theta = 2\angle CAB \rightarrow \angle CAB = \frac{\theta}{2} = \frac{\frac{2\pi}{n}}{2} = \frac{\pi}{n}$$

$$\rightarrow \angle A = \frac{\pi}{n}$$

$$\sin(A) = \frac{BC}{AC}$$

$$\rightarrow \sin\left(\frac{\pi}{n}\right) = \frac{r}{R-r}$$

$$\rightarrow r = \sin\left(\frac{\pi}{n}\right)(R-r)$$

$$= R\sin\frac{\pi}{n} - r\sin\frac{\pi}{n}$$

$$\rightarrow r + r\sin\frac{\pi}{n} = R\sin\frac{\pi}{n}$$

$$\rightarrow r\left(1+\sin\frac{\pi}{n}\right) = R\sin\frac{\pi}{n}$$

$$\rightarrow r = \frac{R\sin\frac{\pi}{n}}{\left(1+\sin\frac{\pi}{n}\right)}$$